Literature-implied answer for a real $p = 3/2$ subcase

arXiv:2012.02016 Question 5 via arXiv:2207.07938

Status. `literature_implied_answer (partial real-scalar variant)`. This is not a new solution of the invariant-subspace problem and not a solution of the complex $1 < p < 2$ case.

Source question. Grivaux–Matheron–Menet, *Does a typical ℓ_p-space contraction have a non-trivial invariant subspace?*, arXiv:2012.02016, Section 7, Question 5, asks:

$$\begin{aligned} &X = \ell_p,\ 1 < p < 2,\ \text{or } X = c_0 : \\ &\quad \text{for every fixed } M > 1,\ \text{is } (MT)^* \text{ hypercyclic for typical} \\ &\quad T \in (\mathcal{B}_1(X), \mathrm{SOT})? \end{aligned}$$

For the ℓ_p case it also asks whether, at least, the typical SOT contraction has no eigenvalue.

Supporting theorem. Grivaux–Matheron–Menet, *Generic properties of l_p-contractions and similar operator topologies*, arXiv:2207.07938, Theorem C (restated as Theorem `toutcapourca`) proves that, for real ℓ_p with $p = 3$ or $p = 3/2$, all topologies on $\mathcal{B}_1(\ell_p)$ lying between WOT and SOT* are similar. In particular, SOT and SOT* have the same comeager subsets on real $\ell_{3/2}$.

Claim. *Let X be real $\ell_{3/2}$. For every fixed $M > 1$, a typical $T \in (\mathcal{B}_1(X), \mathrm{SOT})$ is such that $(MT)^*$ is hypercyclic. Consequently, a typical SOT contraction on real $\ell_{3/2}$ has no eigenvalue.*

Proof. For $1 < p < \infty$, arXiv:2012.02016 records in Proposition `monsteraide` that, for every $M > 1$,

$$\mathcal{H}_M = \{T \in \mathcal{B}_1(\ell_p) : (MT)^* \text{ is hypercyclic and } \|T\| = 1\}$$

is dense G_δ in $(\mathcal{B}_1(\ell_p), \mathrm{SOT}^*)$. The proof is the standard SOT/SOT* hypercyclic-density proof and is field-insensitive, so it applies to the real scalar space as well.

Now take $X = \ell_{3/2}$ over $\mathbb{R}$. By arXiv:2207.07938, Theorem C, SOT and SOT* are similar on $\mathcal{B}_1(X)$; hence they have the same comeager sets. Therefore $\mathcal{H}_M$, already SOT*-comeager, is also SOT-comeager.

Finally, if $S = (MT)^*$ is hypercyclic, then $S^* = MT$ has no eigenvalue. Thus T has no eigenvalue. $\square$

Scope. This answers only the real $p = 3/2$ scalar variant/subcase of the hypercyclicity part of Question 5. It does not answer the original complex $1 < p < 2$ problem, the c_0 case, the question whether SOT* comeagerness transfers to SOT for all $1 < p < 2$, or the main invariant-subspace question.

Search evidence. The run indexes were searched for arXiv:2012.02016, the title keywords, SOT/SOT*, hypercyclicity, and invariant-subspace phrases. The only close prior durable hit was the attempt note for arXiv:2207.07938. Web searches on 2026-07-03 for the exact 2012 title and the $p < 2$ hypercyclic/no-eigenvalue phrases found no newer complex-case resolution.

References.

- S. Grivaux, E. Matheron, Q. Menet, *Does a typical ℓ_p-space contraction have a non-trivial invariant subspace?*, arXiv:2012.02016.

- S. Grivaux, E. Matheron, Q. Menet, *Generic properties of l_p-contractions and similar operator topologies*, arXiv:2207.07938.